

Monsoon Semester, 2021, Faculty of Technology, CEPT University

Code	SD4004	Course	Design of Steel Structures		
Typology	Studio	Credits	12	GPA/NonGPA	GPA

Long Description :

The course will prepare students to analyze and design steel structures independently using “Limit State Design Method” including limit state of strength and serviceability. The design approach will be based on the latest Indian Standard Code of Practice for general construction (IS 800:2007). The course will cover all important components such as material specifications, loading standards, elementary design and connections of structural members for designing steel structures. The course will provide relevant material properties of different types of steel and design considerations. Design of gantry girder with reference to industrial applications will also be covered including fatigue design. The course aims to develop professional competencies in the design and construction of steel members through critical thinking led by discussions and analyses of various practical steel structural design problems. The course will enable fresh BE students to work individually on given project with thorough understanding of architectural drawings, preparing structural systems using latest Indian Standard Code of Practice for general construction (IS800:2007). Students will apply their knowledge from strength of materials and structural analysis for further understanding between analysis, selection and design of structural steel elements and their connections. Students will also learn the use of structural software for structural analysis. The course will also cover the structural design of industrial facilities in a systematic manner, starting with loads, load combinations, and different types of structural systems and framing concepts. The course will include analysis and design procedure of steel members under axial tension, roofing / flooring systems, compression members, built-up compression members, laterally supported and laterally unsupported beams, beam-columns and column bases which will transfer load from super structures to foundation through pedestal and finally to underneath soil. The course will cover both bolted and welded connections. Eccentric connections due to different plane of loading for both bolt and weld will be discussed. Students will gain competency in analytical theories, structural systems and detailing structural drawings with main emphasis on the understanding of the basic behavior of steel structures. Students will translate architectural drawing into structural system and overlap it to get thorough understanding. The studio will also invite architects to give inputs on chosen structural systems. The course will be supported by numerous assignments and design project work to understand the defined concepts. Students will prepare well documented report and portfolio along with structural drawings at the end to communicate their conclusions in writing and also learn the presentation skills for the same.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will learn to read architectural drawings – plans, elevations and sections of the given project. Students will have an overview of structural steel elements and their behavior.
- Students will gain knowledge on current codes of practice and guidelines and how these specifications are used in the solution of practical design problems
- Students will learn to design steel structures and understand the behavior of gantry girders with reference to industrial applications including fatigue design
- Students will understand the concept and behavior of columns subjected to axial force and moments along with lateral torsional buckling and design methodology for such columns
- Students will prepare detailed structural drawings to execute on site

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Allocation of the project, introduction to steel structures, types of cross – sections, physical properties, types of structural steel elements and their connections, design philosophy and limit state method of design
Week 2	Preparing structural system of the project
Week 3	Preparing structural system of the project
Week 4	Assessment of loads on various structural elements
Week 5	Analysis and design of roofing/ flooring system
Week 6	Analysis and design of beams
Week 7	Analysis and design of gantry girders
Week 8	mid sem jury
Week 9	Analysis and design of columns
Week 10	Design of column bases and bracings
Week 11	Analysis and design of connections
Week 12	Analysis and design of pedestal
Week 13	Analysis and design of tie beams
Week 14	Analysis and design of footings
Week 15	Detailing of structural elements
Week 16	Jury/final exam; Portfolio prep; Final portfolio examination

Monsoon Semester, 2021, Faculty of Technology, CEPT University

Code	SD4611	Course	Earthquake Engineering		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Engineering Seismology is the field mainly explored by the seismologists. However, the effect of earthquakes on the engineering structures needs a link between engineering seismology and earthquake structural engineering. This course will mainly focus on understanding the seismological aspects on structures and will formulate the link with engineering aspects of structures. In second part of the course, conceptual understanding of earthquake forces and reaction to those forces by the structure will be at core. This will ensure the objective of understanding the connection between seismology, earthquake forces and reaction of earthquake forces. This course aims to connect the missing link between engineering seismology and structural earthquake engineering. In initial lectures, seismological understanding will be discussed in detail. This understanding will be built-up with quantification of the earthquake effects. Interesting discussions will be happening for co-relationship between magnitude, intensity and ground acceleration. This will create the base for the conceptual understanding of reaction of building to earthquakes. In next set of lectures, force, mass and acceleration will be connected with stiffness and natural period. This will be backed up with discussions on inertia, damping and stiffness. Last set of lectures will be mainly focused on codal provisions and detailed discussions will be on important clauses of IS:1893-2016. Above lectures will create scope for interesting assignments, models, case studies and compilation of research papers.

Learning Outcomes :

After completing the Courses, the student will be able to :

- After completion of this course, students will link seismological aspects with engineering aspects of structures.
- Students will be ready for the Studio II, because, the fundamentals of the codal provisions will be discussed on detail, which will be applied in Studio II

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Monsoon Semester, 2021, Faculty of Technology, CEPT University

Code	SD4612	Course	Evaluation of Failures, Repair and Rehabilitation of Structures		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Recent years it has been observed that many structures show the signs of distress or may have collapsed during the early stage of the service life. The causes are attributed to many external man caused factors, such as defective design, faulty construction and poor maintenance. This course is designed keeping these causes in mind and highlight the role of an Architect and an Engineer at Design, Construction and Maintenance stage. It discusses about emphasizing good engineering practices that are to be followed at the design and construction levels to achieve desired performance during the service life of such structures. The sessions are planned from defining the construction failure and its historical reference and present causes in the buildings. Then the discussion will be on analyzing and evaluating the failures through visual inspection, nondestructive testing and other scientific methods. After the required investigation methodology is discussed, suggestive methods to be adopted for repair and strengthening the structures will be discussed. Students will take case studies of failures and while discussing the cause of failure, will also give suggestive methods to repair and restore the same.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will be able to know the types and causes of failures in structures
- Students will be able to identify the repair techniques and materials
- Students will be able to evaluate the cause of the failure and will be able to justify the suggestive repair technique needed to be applied on the structures.

Monsoon Semester, 2021, Faculty of Technology, CEPT University

Code	SD4613	Course	Advanced Structural Analysis		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

To understand the behavior of different types of framed structures through advanced matrix methods of analysis, which are used in the popular computer software. Initially to learn the analysis of structures, having lesser degree of freedom (dof), with manual calculations; and thereafter its application to structures having large dof with help of computers. To develop standard procedures to incorporate any type/size of structure with all possible loads/load combinations. Stiffness and flexibility methods of analysis will be discussed. To define the structural model and loads prior to analysis [pre-processing] and also interpretation of analysis results [post-processing]. To learn the application of the method to real-life problems/conditions in practice. To validate/verify the hand-calculations with computer software for several examples. The main emphasis shall be on the behavior, analysis methodology and interpretation of analysis results for large dof structures generally encountered day-to-day in structural engineering design practice; which is a primary pre-requisite to elementary designs.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will be able to understand a general behavior of structures and revise/refresh the pre-requisites such as matrix algebra.
- Students will be able to learn the difference in approach of the two basic matrix methods of analysis and solve elementary examples
- Students will be able to learn analyzing simple lesser dof structures from basic concepts, with hand calculations
- Students will be able to understand the difference between system approach and member approach for different framed structures, and extend their basic understanding to analysis of higher dof structures. They will understand the human limitation, and appreciate the aid of computers.
- Students will be able to learn the application of the methods to some real-life conditions in practice. Students will be able to understand the basic difference in the behavior of a framed structure and continuum structure and the approach to their analysis.

Winter Semester, 2021, Faculty of Planning, CEPT University

Code	W21FP002	Course	Sustainable Practices in Built Environment		
Typology	-	Credits	2	GPA/NonGPA	GPA

Long Description :

Built environment are the spaces planned, designed and constructed by humans to live and work. These may be residential, commercial, industrial, institutional, parks and gardens etc. A sustainable built environment imitates nature and is circular. It uses low-carbon, low-impact, eco-friendly materials, efficient use of resources and follows the principles of reduce, reuse, and recycle. Deploying these principals preserves and protects natural resources by reducing wastes, energy consumption, emissions, pollution; and make cities sustainable and resilient for its citizens. It reduces per capita environment impact of the cities and also improves the living environment which has positive impact on human health. The course would focus on various aspects of sustainability which should be included in the built environment. The course would include theory and practice, the learning is proposed to be strengthened with presentation and discussion of case studies and field visits.

Learning Outcomes :

After completing the Courses, the student will be able to :

- The students would learn about the regulations, guidelines and certification procedures promoting sustainable practices. They would learn about different dimensions of sustainability in built environment. They would know about how sustainable practices are planned, designed, implemented and maintained in the built environment. Through the practice exercise, the students would be able to gain knowledge about implementation of such sustainable practices.

Daily Plan & Assessment :

Days	Exercise Description
Day 1	Introduction to the course, Discussion on SDGs, Green building and certification, introduction to exercise
Day 2	Discussion on sustainable practices in different thematic areas by experts, visit to green building
Day 3	Redevelopment and rehabilitation and site visit
Day 4	Provision of sustainable practices in planning regulations, policies, and guidelines, Financial Sustainability
Day 5	Preparation, presentation and submission of exercise document & discussion

Spring Semester, 2022, Faculty of Technology, CEPT University

Code	SD4006	Course	Design of Reinforced Concrete Structures		
Typology	Studio	Credits	12	GPA/NonGPA	GPA

Long Description :

This studio focuses on design of structures with reinforced concrete. Discussions will range from master designers, iconic buildings and related design philosophy and structural load transfer to the modern futuristic buildings. Students will be assigned individual projects along with the problem statement and they will develop the structural system considering the gravity loads only at the initial stage. Students will also prepare the physical models which will then be tested on the shake table to understand the different structural modes, follies in design and methods to rectify the same. Individual component design will be dealt during master class sessions and weekly exercises will be formed to get a better view on designing them under varying load considerations. The variations in the structural system will be discussed and students will apply the same. Students will actually be able to interpret the difference in behavior of the buildings with the systems adopted and designed with gravity loads and lateral loads. This procedure of arriving at an optimized structural system will be well documented and results will be explained in the report. Depending on the availability of laboratory, beams and cubes will be cast and will be loaded till failure through which the failure modes will bring in the relevance of reinforcement placement in a structure. The complexity of loads will be introduced at the later stage. The structure will have to be finally designed for static as well as for dynamic loads. The final output will be in form of a report, files in software and a detailed drawing for a floor, a frame with shear wall and a foundation.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will be able to develop alternatives for the structural system for a given space.
- Students will apply the principles of wind and earthquake engineering and limit state theory in designing a structure.
- Students will learn to mitigate the effects of lateral loads and will apply an appropriate solution to the periodical challenges faced while designing a building.
- Students will learn to communicate the final design in the form of structural drawings that are required at site.

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	[ON] –Mon, Wed, Thu] Introduction to reinforced concrete structures. Discussion about forms of structures in nature that have inspired to create the modern buildings. Discussion on the design philosophy of master builders. Historic development of theories of designing structures. Assign the individual project and discussion on the problem statement Students will be able to interpret the load acting on a structure and its transfer path. Students will be able to learn the basics of the development of modern theory.

Week 2	[ONl –Mon, Wed, Thu] Discussions will be based on the basis of developing a structural system for the given project. Criteria for load consideration, sizing and developing a load transfer path will be discussed. Loads will be restricted to gravity loads. Students will prepare various structural systems on tracing paper and document the evolution of the final system. Students will learn to develop a structural system for the project assigned.
Week 3	[ONl –Mon, Wed, Thu] Introduction to limit state theory and related analysis methods will be discussed. The application of these methods to slabs and beam design shall be covered. Simultaneously, the detailing aspect of the same shall be introduced along with the relevant codal provisions. Based on final structural system, students will then detail the floor slab.
Week 4	[ONl –Mon, Wed, Thu] Introduction to limit state theory and related analysis methods will be discussed. The application of these methods to slabs and beam design shall be covered. Simultaneously, the detailing aspect of the same shall be introduced along with the relevant codal provisions. Based on final structural system, students will then detail the floor slab. Students will be able to analyze and detail slabs. Assessment will be based on assignments related to analysis of the frame.
Week 5	[ONl –Mon, Wed, Thu] Students will generate the software model for the problem assigned and finalized structural system. Students will then run the model and extract the results for beams and columns. They will study the pattern related to the final forces. Students will do the required static checks done to establish the validity of the model. Manual results will be verified with the software results. Students will interpret the pattern of the force transfer. Students will be able to extract the forces and will be then able to design and detail the selected frame. Assessment based on the detail of final floor slab and interpretation of results.
Week 6	[OCp – Mon, Wed, Thu] Sizing of the columns will be discussed with respect to loads coming on them. Analysis of the columns will be discussed. Students will be able to decide the final size of columns and will document the same in terms of size and reinforcement required.

Week 7	[OCp – Mon, Wed, Thu] This week is to be planned at CEPT. Introducing the lateral loads in the system. Briefing of IS 1893 and discussion of effects of earthquakes across the world. Understanding the behavioral difference under lateral loads. Briefing about wind loads and methodology to apply the same on the building. Students will prepare models and test on shake table when they are at CEPT. Students will be able to interpret clauses of IS 1893. Students will study effect of earthquakes by considering few examples and presenting the same. Students will study irregularities and arrive at the final structural system after analyzing different combinations. Students will document their observations related to the models failures. Students will be able to document the differences by analyzing the building under gravity and lateral loads by using small examples and laboratory testing. Research papers will be discussed and students will write a summary of the topic of the problem statement allotted.
Week 8	[OCp – Mon, Wed, Thu] This week will be also at CEPT. Application of the loads in the model continues along with the final sizing with respect to reduced moment of inertia in building. Application of dynamic analysis. Students will give be able to give final differences with respect to sizes with gravity loads, lateral loads, reduced moment of inertia and dynamic analysis.
Week 9	[ONl –Mon, Wed, Thu] Introducing shear walls in the building and reworking the sizes of the columns and beams. Bringing building safe under all the required checks. Students will interpret the results pertaining to inclusion of shear wall in the building. Students will then optimize the system.
Week 10	[ONl –Mon, Wed, Thu] Performing analysis, selecting a frame for the final results. Introduction to the clauses of IS 13920. Designing a shear wall Students will be able to apply the sizing and detailing aspects in the buildings. Students will learn to interpret the plate stresses and will learn to design a shear wall.
Week 11	[ONl –Mon, Wed, Thu] Discussion on foundation design and detailing. Concept of grouping of columns and foundation will be discussed.
Week 12	[ONl –Mon, Wed, Thu] Detailing a frame, shear wall and a foundation. Preparing for the final jury. Students will learn to document the necessary details from the work done till date.
Week 13	[ONl –Mon, Wed, Thu] Portfolio preparation Students will document the exercises and prepare the final structural detailing drawings based on the suggestions in the jury.
Week 14	[ONl –Mon, Wed, Thu] Portfolio preparation Students will document the exercises and prepare the final structural detailing drawings based on the suggestions in the jury.

Week 15	[ONl –Mon, Wed, Thu] Portfolio preparation Students will document the exercises and prepare the final structural detailing drawings based on the suggestions in the jury.
Week 16	[ONl –Mon, Wed, Thu] Portfolio preparation Students will document the exercises and prepare the final structural detailing drawings based on the suggestions in the jury.

Spring Semester, 2022, Faculty of Technology, CEPT University

Code	SD4614	Course	Wind Engineering		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Wind speed vary randomly both in time and space and hence assessment of wind loads and its response predictions are very important in the design of structures. Wind analysis governs the design of tall buildings, steel structures, slender towers and structures situated in hurricane prone regions. Buildings and their components are required to be designed to withstand code-specific wind loads, hence study of codal provisions is vital to design wind force-resisting system for a building. The course includes study of basic terminologies based on IS 875 (part-3)-2015 and understanding different codal provisions to calculate wind pressure using two different methods, i.e. static analysis and dynamic wind response and their requirements. Wind load calculation for different types of RCC and steel structures will be explained using various examples. Along with theoretical knowledge regarding wind load calculations, the course also includes practical application of wind loads in structural analysis of various types of buildings using STAAD software. A specialized project will be allocated to each student to properly study wind pattern for a particular structure along with project specific discussions to include wind-resisting components in a structure to control wind induced deflections.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Interpret the codal provisions of IS-875 (Part-3)-2015
- Calculate wind pressure for all types of structures
- Perform wind analysis manually as well as through software

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction: Introduction to the concept of wind analysis, it's requirement and understanding of basic terminologies based on IS 875 (part-3)-2015 such as solidity ratio, shielding effect, basic wind speed, velocity profile, effective frontal area, gust, fetch length, terrain category etc.; serviceability and design load combinations for wind analysis; Permissible deflection limits
Week 2	Introduction: Introduction to the concept of wind analysis, it's requirement and understanding of basic terminologies based on IS 875 (part-3)-2015 such as solidity ratio, shielding effect, basic wind speed, velocity profile, effective frontal area, gust, fetch length, terrain category etc.; serviceability and design load combinations for wind analysis; Permissible deflection limits Introduction: Introduction to the concept of wind analysis, it's requirement and understanding of basic terminologies based on IS 875 (part-3)-2015 such as solidity ratio, shielding effect, basic wind speed, velocity profile, effective frontal area, gust, fetch length, terrain category etc.; serviceability and design load combinations for wind analysis; Permissible deflection limits

Week 3	Static wind analysis: Step by step understating of wind pressure calculation using static wind analysis method from IS 875 (Part-3)-2015; Study of force coefficient tables and basic understanding of wind force directions (pressure and suction) on a structure for along and across wind analysis
Week 4	Static wind analysis: Study of wind pressure calculation on RCC structures using static wind analysis method with an example. Change in wind pressure due to different shape and size will be demonstrated through solved examples in class.
Week 5	Static wind analysis: Study of wind pressure calculation on different types of steel structures using static wind analysis method with few examples of cladding conditions and varying slope of the roofs.
Week 6	Dynamic wind analysis: Introduction to the concept of dynamic wind analysis, it's requirement and understanding of basic terminologies based on IS 875 (part-3) such as Interference effect, motion due to vortex shedding, sub and super critical flow, along and across wind response etc.
Week 7	Dynamic wind analysis: Step by step understating of wind pressure calculation using dynamic wind response from IS 875 (Part-3)-2015
Week 8	Dynamic wind analysis: Study of wind pressure calculation on RCC structure using dynamic wind response with an example
Week 9	expert lecture
Week 10	Wind analysis in software: Understanding wind load application on RCC and steel structures in ETABS/STAAD software with examples (Or expert lecture)
Week 11	Wind analysis in software: Understanding wind load application on RCC and steel structures in ETABS/STAAD software with examples (Or expert lecture)
Week 12	Specialized project: Allocation of a specific project to each student to carry out wind calculations manually and verify the same with software results. The results to be interpreted and the wind induced deflection should be controlled within prescribed limits.
Week 13	Specialized project: Allocation of a specific project to each student to carry out wind calculations manually and verify the same with software results. The results to be interpreted and the wind induced deflection should be controlled within prescribed limits.
Week 14	Specialized project: Allocation of a specific project to each student to carry out wind calculations manually and verify the same with software results. The results to be interpreted and the wind induced deflection should be controlled within prescribed limits.

Spring Semester, 2022, Faculty of Technology, CEPT University

Code	SD4615	Course	Nonlinear Analysis of Structures		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

All structures are generally designed for seismic resistance using elastic approach, which is a popular and conventional method. It is assumed that a structure behaves elastically to earthquakes where in the stresses in the materials are within the linear elastic range along with constant stiffness matrix. In reality, structures experience substantial inelastic deformations during massive earthquakes which occur due to geometric nonlinearity, material nonlinearity and contact and as a result the stiffness matrix is not constant throughout load application. Hence, nonlinear analysis is vital to capture realistic behavior of a structure under seismic effects. Nonlinear analysis does play an important role in design of new structure and retrofit an existing structure. The development of new codes for earthquake-resistant structures also stress on nonlinear analysis of buildings for better performance of structures, when subjected to seismic forces. To capture a structure's inelastic behavior (inelastic deformations), a structural engineer has to create a model in software, incorporate various non-linearity and conduct simulations. The advancement of computational tools has led to the proliferation of software packages in the design industry which has facilitated the designers to make decisions on the required performance criteria of the structure. In order to achieve this, the software package should be thoroughly understood with all input parameters and options available to carry out meaningful nonlinear analysis. An accurate mathematical model of the structure in software is must to conduct nonlinear analysis and gain confidence in the evaluation process. The course will cover basic understanding of nonlinear modeling and analysis of structures when subjected to dynamic loadings, focusing on the seismic response of structures. Introduction and background of linear and nonlinear analysis will be briefed. Recent developments and standards to carry out nonlinear analysis will be appraised. The course will introduce nonlinear analysis software with modeling examples. Students will be taught creating 3D model in software, assigning properties, loads and load combinations as per Indian Standards, analysis and design of structure using static and dynamic analysis, followed by nonlinear static pushover analysis, assigning nonlinear properties, obtaining capacity curve and performance point. Introduction to Time history analysis and obtaining energy output will be covered.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Analyze and design RC building using static analysis and dynamic analysis
- Understand the importance of nonlinear analysis and software with modeling examples
- Hands on experience to carry out static non linear pushover analysis and interpret the results
- Hands on experience to carry out time history analysis and interpret the results

Spring Semester, 2022, Faculty of Technology, CEPT University

Code	SD4616	Course	Advanced Geotechnical Engineering		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Objectives of the course are to know about various field and laboratory investigation techniques, to interpret the observations of geotechnical investigations. The basic principles of shallow and deep foundation design with reference to site investigation, soil classification and review of index properties will be covered in the class. Discussions will be done on various experimental tests required to understand the various parameters of soil. This course will also cover the methods of determining the bearing capacity and settlement calculations. Methods of ground improvement and reclamation techniques and applications for the same in the given condition of site shall be discussed. After considering the liquefaction potential of the soil, criteria for final tender preparation will be taught. Students will learn to read and interpret the soil report and apply geotechnical engineering principles for the design of foundation structures. Students will work on actual reports of the project also learn to do the sizing of shallow isolated, combined and raft foundations based on the given values. The concept of pile capacity under individual and group of piles will be covered at the later stage. Certain soil tests would also be demonstrated at KBM laboratory.

Learning Outcomes :

After completing the Courses, the student will be able to :

- After completing the Course students will be able to decide the extend of field and laboratory geotechnical investigations required for any construction project and develop the specifications for the same.
- Students will also be able to interpret the observations of geotechnical investigations and apply the same for the sizing of foundation for structures
- Students will also be assessing the liquefaction potential and apply the appropriate ground improvement techniques, in case required.

Spring Semester, 2022, Faculty of Technology, CEPT University

Code	CT2609	Course	Project Estimation		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Quantity Surveying and Project Estimation is one of the essential skill for handling any civil engineering project. This subject introduces students to units of measurements, bills of materials, different methods of measurement for quantity of materials and construction related items and rate analysis of main construction activities. This subject aims to make students capable of performing basis functions in Quantity Surveying Practice. It will be done by taking out quantities for smaller independent structures such as security cabin, compound wall, manholes etc. Students will learn methods and process of bills of quantities, detailed estimate, specifications of materials and analyze rates. Student will learn process of doing rate analysis of basic construction items such as earthwork, brick work, concrete, formwork, reinforcement, flooring, wall finishing etc. It will develop students' skills, techniques and abilities with regard to the role of the quantity surveyor which are very essential in field of construction. Students will be able to prepare estimates for projects by learning all necessary skills during the course. This course will have a two hour tutorial session on the same day from 5:00 pm to 7:00 pm.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Student will be able to select mode of measurement for various building items
- Student will be able to take out quantities for building project activities from the drawings
- Student will be able to analyze and derive unit rates for construction activities
- Student will be able to prepare detail estimate from quantities and unit rates
- Student will be able to develop constants to be used for future projects by summarizing data

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Units and mode of measurement
Week 2	Quantity take off
Week 3	Quantity take off
Week 4	Quantity take off
Week 5	Quantity take off
Week 6	Quantity take off
Week 7	Quantity take off, Mid term Test
Week 8	Specifications
Week 9	Specifications
Week 10	Rate Analysis

Week 11	Rate Analysis
Week 12	Rate Analysis, Term end test

Summer Semester, 2022, Faculty of Architecture, CEPT University

Code	S22FA006	Course	Parametric Design using Grasshopper3D		
Typology	-	Credits	2	GPA/NonGPA	GPA

Long Description :

The term parametric design has been in the field of architecture for more than 60 years but recently during the last couple of decades it has come in limelight and people are thriving to pursue it every day. But do we actually know what Parametric Design is and who started it? Or how do we become a parametric thinker/designer with advanced computational tools? The workshop on Parametric Design answers these questions by learning and adapting the parametric way of designing. In these sessions, participants will learn how to approach designs parametrically, using a combination of geometric and mathematical tools inside of David Rutten's Grasshopper 3D, that will bring their design ideas to life. Each day will be focused on a specific topic like - 1. Creating Virtual Magnets - Topic on creating attractor objects to manipulate the design and make kinetic facades. 2. Manipulating Images - Make designs and facades by transforming images. 3. Working with Graphs - Creating smooth undulating surfaces using Parametric Graphs. 4. Decoding Pavilions & Non-Planar Surfaces - Developing experimental forms used in pavilions and temporary structures.

Learning Outcomes :

After completing the Courses, the student will be able to :

- By the end of the course, participants will have learnt the principles and practices of Parametric Design by employing computational design tools. They will learn to manipulate scripts, automate their designs and create an iterative design loop in the computational world. Participant will also get a peek into how designers & architects are scripting or coding their designs. Along with this, inclined B. Arch participants will gain insights into postgraduate level studies in Computational / Parametric / Advanced Architecture.

Daily Plan & Assessment :

Days	Exercise Description
Day 1	1. In this module, students will learn how to create attractor objects to manipulate their designs and make kinetic facades. 2. Initial introduction to Grasshopper and its interface. 3. How scripts and components function in Grasshopper. 4. Re-creating examples like – Aqua Tower by Studio Gang, Façade of Al Bahr Tower by Aedas Architect and AU Office & Exhibition Space by Archi Union Architects.
Day 2	1. In this module, students will learn how architectural facades are made using images. 2. Recap of last class. 3. Mapping colors to numbers from Images. 4. Data manipulation with by understanding Data Sets. 5. Re-creating examples like - The Nelson Mandela Monument by Marco Cianfanelli and The MaoHaus by AntiStatics Architecture.

Day 3	1. In this module, students will learn to design smooth undulating surfaces using parametric graphs. 2. Understanding how parametric graphs like Sine, Gaussian, Perlin and other work. 3. Transforming the numerical data from these graphs to be used for architectural surfaces. 4. Re-creating examples like - The Wave Structure Façade by Scott Brownrigg Architects, Bodegas Ysios Winery by Ar. Santiago Calatrava and Regal Shoes by Nudes Architects
Day 4	1. Taking the knowledge from last 3 class in grasshopper, students will make complex pavilions by decoding their design process. 2. Developing vectors for undulating or non-planar surfaces. 3. Creating patterns on curvilinear surfaces by evaluating these surfaces. 4. Re-creating examples like - UGUNS Pavilion by Didzis Jaunzems Architects, Digfabmtly 1.0 by Arquidromo and Lincoln Park Zoo Pavilion by Studio Gang.

Summer Semester, 2022, Faculty of Design, CEPT University

Code	S22FD012	Course	Braids of Joy: Create, Deconstruct, Mend, Reconstruct (using Ply-Split Braiding)		
Typology	-	Credits	3	GPA/NonGPA	GPA

Long Description :

The craft of Camel Belt making utilizing the technique of Ply-Split Braiding comes from the western desert region of Rajasthan and Gujarat. It is an age-old craft and uses a lesser known, non-loom technique. The course offers an opportunity to understand the craft through a practice based approach. The specificity of the materials would be shared to equip and empower the participants to make the right choice while visiting the market for material procurement. The right choice, to contribute as part of their contribution towards the whole. Using the collected materials, the course would begin with all the participants working on a huge form, collectively. The understanding of the various structures and the nuances of the technique would be learnt on the go. The created outcome would then be deconstructed with the awareness of its reconstruction with the aid of mending, to save the fragments from disintegrating, fraying. This would lead to the reassembling of the fragments to present another view, another story, another tactile experience: A new outcome. willingness to learn with an open heart

Learning Outcomes :

After completing the Courses, the student will be able to :

- The learning outcome is going to be a huge tactile form, representative of the collective. The huge form would be collectively visualized with scope for all individual identities to emerge and shine. It is going to be a process driven outcome. The takeaways are going to be rich, colourful experiences and the mended parts of the whole. Participants would be encouraged to document the process and note down observations of their learnings for future reference. And for sharing the joy of braids, ahead, with the ones who are willing to learn and share.

Daily Plan & Assessment :

Days	Exercise Description
Day 1	Basics of the technique along with the specificity of the materials to be introduced to the participants. This is done in order to equip the students to explore the market and bring back materials along, as per their liking; appropriate for use in the technique of Ply-Split Braiding. Each participant would be visiting the market on day 1, Saturday, bringing back the material for further discussion. A visit may again be made on Sunday if need be.
Day 2	Discussion on the collected materials, material preparation for further use, visualizing and arriving at the one outcome, form that would be conceived and worked on, collectively. The final deliverable, the shape/ form/ colour/ texture etc. is subject to conversations and how the physical material unfolds in real time and space.
Day 3	Laying the foundation of the visualized outcome. This is going to be a team effort with equal participation.
Day 4	Work on the final form to continue

Day 5	Work on the final form to continue with the awareness that it is to be deconstructed at the end of its creation.
Day 6	Work on the final form to continue with the awareness that it is to be deconstructed at the end of its creation. The deconstructed fragments are to be saved through mending. Converting each fragment into a stable entity, on its own.
Day 7	The completed final outcome to be deconstructed with awareness for reconstruction. The fragments are to be mended as soon as deconstruction takes place.
Day 8	The act of mending to save the fragments to continue on Saturday. Followed by reassembling the fragments to present another view, another story, another tactile experience: A new outcome. Participants would be encouraged to document the process and note down observations of their learnings, individually.

Monsoon Semester, 2022, Faculty of Technology, CEPT University

Code	SD4003	Course	Design of Industrial Structures		
Typology	Studio	Credits	12	GPA/NonGPA	GPA

Long Description :

The studio focuses on the analysis, design and detail of heavy-duty industrial structures. Industries such as cement industry, refineries, sugar industry, power plants etc. have a requirement of various categories of structures such as chimneys and silos and other industrial buildings and units. When the theory of analysis of each structure is different, it is essential to know all the required provisions for designing such complex problems. Discussion on important industrial structures by showing the requirements of existing power plant, sugar, cement, oil refinery, paper and paint industries will be undertaken during the course of this studio. Discussions will happen on technical specifications, process requirements and its structural planning to fulfil the desired functions. Identifying essential features that play an important part in forming any industry such as Silos, Chimneys, preheaters, Industrial sheds, pipe racks and conveyor belt systems from the live projects discussed will be a part of the course. Major components of the industrial structures i.e., silos, chimneys and other functional units will be covered in detail. Therefore, such projects will be undertaken and structural design of the project in part or whole will be accomplished by the students. The outputs will be presented in form of design reports and detail drawings. Silos are the integral part of many industries. Understanding the methods/procedures for analysing and designing of Silos with various configurations will be done by giving exposure to existing codes IS 4995 regulations for these structures. Silos design is sometimes also governed by international norms. The provisions adopted in American ACI 313 and Eurocode 2005 will also be discussed. Problem solving exercises will be given to solve various conditions of loads. With the strict norms from pollution control board, chimneys are regarded as one of the important structures for large industries. Understanding the methods/procedures for analysing and designing of Chimneys will be done by discussing the norms given in IS 4998. The combined effect of temperature, earthquake and wind loads will be shown in the class and student will be then working on the assigned problem. Any basic industry has standard requirements in terms of building and other functional units. Discussions will be done based on existing codes IS 800, IS875 and system formation for industrial sheds, preheaters, conveyor system, machine foundation and pipe rack system will be discussed.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Students will be appraised with the functional requirements of different existing industries.
- Students will be well equipped with regulatory norms and design methodologies required for special structures.
- Students will be well equipped with analysis of the assigned structures.
- Students will be independently able to design the assigned industry. Students will be able to prepare DBR that is required during proof checking submission. Students will deliver drawings those are to be released on site for execution.

Weekly Plan & Assessment :

Weeks	Exercise Description
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Week 1	Discussion on important industrial structures by showing the requirements of existing power plant, sugar, cement, oil refinery, paper and paint industries etc. Discussions will happen on technical specifications, process requirements and its structural planning to fulfil the desired functions. Identifying essential features that play an important part in forming any industry such as Silos, Chimneys, preheaters, Industrial sheds, pipe racks, conveyor belt systems from the live projects discussed.
Week 2	Going through the provisions, documenting and understanding the behaviour of different types of silos. Calculation of pressures for the problems assigned as per American, European and Indian codes. Codal provisions and various gravity load calculations for chimneys will be discussed.
Week 3	Understanding codal provisions and calculation of the pressure in silos for the problems assigned as per American, European and Indian codes. Also, student will identify their behaviour of silo under the various filling & emptying loading condition for their individual problem statement. Discussion of codal provisions in chimney along with understanding and performing along wind load calculations.
Week 4	Comparison and study for pressure calculations from various codal provisions in silos, along with the design provisions for shell according to the governing pressures. Discussion of codal provisions and along & across wind load calculations for chimneys will be undertaken.
Week 5	Analysis and design of ring beams, cone, top slab and base slab for various types of silos will be done. Students will interpret the analysis results from software and will do the design manually. Detailing aspects as per IS 13920 and IS11682 will be discussed. Discussion of earthquake and temperature load calculations for chimneys will be undertaken.
Week 6	Design for lateral load in silos along with special design and detailing aspects at the opening. Discussion of codal provisions for chimneys along with the design and detailing of shell and foundation. In chimney design and detailing of important elements like corbel, ring beam, cage ladder etc. will also be discussed.
Week 7	For both Silo & Chimney problem statement, completion of the design and detailing for all the components covered will be completed and students will undertake the compilation of design calculations and drawings.
Week 8	Submission of the complete analysis, design and detailing part of chimneys and silos along with the Mid semester Jury and exams. Parallel to mid term exam students will undertake the revision of the basic concepts of steel structure design.

Week 9	In miscellaneous heavy-duty structures, as each structure has different functional requirements, this will be discussed in class and students will be assigned the live projects. Presentation of understanding regarding the project by the students, along with preparation of DBR (Design based report).
Week 10	Formation, discussion of the structural systems for the individual projects. Discussion and completion of gravity load calculations and discussion for lateral loads (wind loads and earthquake loads) will be done. Structural system for lateral loads will also be covered.
Week 11	Completion of gravity & lateral load calculations. Student will start modelling of their projects in the software. Parallely preparation of spreadsheets for load calculations will be undertaken by the students.
Week 12	Finalization of structural system for both gravity and lateral loads along with model completion in software with all the load calculation. Structural analysis results along with their interpretation will be undertaken here.
Week 13	Design parameters for various structural members will be discussed and design as per IS-800 will be covered for various members. Parallely preparation of spreadsheets for member design will be undertaken by the students.
Week 14	Completion of assigned member design both in software and manual design in spreadsheets. Connection design and detailing will be discussed during this week.
Week 15	Completion of the design and detailing of their given scope by the students along with the final report of all design calculations with all the necessary drawings.
Week 16	Final Jury and portfolio grading

Monsoon Semester, 2022, Faculty of Technology, CEPT University

Code	SD4617	Course	Design of Marine Structures		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

The course will give comprehensive understanding on role of ports and their importance in international trade and transport there by acting as a backbone of country's economy. Different types of ports and access to ports, functions, operations and management will be incorporated. Requirements of design of marine structures will be discussed. An overview of port terminal along with its components will be studied. The course will touch upon the historical development of ports and technologies used with description of some famous ports at that course of time. Introduction to Indian ports, their classification, recent developments and growth potential and type of cargo handling will be discussed. An overview of world's busiest ports will be covered. Knowledge of admiralty chart and other important terminologies will be discussed. The course will provide detailed understanding of coastal erosion and various methods of coastal protection. Knowledge of port planning using relevant guidelines along with vessel characteristics will be provided. Thorough understanding of different vessel types based on cargo handled, size, dimension and weight will be provided. Various types of onshore structures along with their structural systems, constructed at different water depths and brief overview of various environmental loads acting on these structures, will be discussed along with the structural action to counteract the encountered loads. The course will cover fundamental principles of wind, wave, tide, current and tsunami characteristics along with design of entrance channel, turning circle, mooring, berthing, ship motion, berthing structures, fenders, bollards, dry docks, slipways, breakwaters, etc. The course will give understanding of modeling and analysis techniques in software, loads and their combinations, design principles, construction methodology and integrity monitoring of RC piled jetty. The course will also provide introduction to breakwaters, their functions, types, failure modes and general design principles. The course will cover different types of materials used for construction in marine environment. Guidelines associated with selection of materials for marine environment, problems associated with the material behavior in marine environment and various protection methods will also be highlighted. The course will be supported by tutorials and design project work to understand the defined concepts.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Learn different loads acting on marine structures including functional and environmental loads
- Analysis and design of RC piled jetty
- Understand vessel characteristics, port characteristics

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction to port and harbors, their importance and classification, Historical development of port structures, Indian ports
Week 2	Coastal protection methods, basic design criteria for port structures
Week 3	Vessel characteristics – types, sizes, cargo carrying capacities, different tonnages and terminologies

Week 4	Ports characteristics – wind, wave, tide, current, tsunami
Week 5	Different loads applicable on RC piled jetty, load combinations, and factors
Week 6	Analysis and Design of RC piled jetty - structural layout
Week 7	mid-term jury
Week 8	Analysis and Design of RC piled jetty
Week 9	Analysis and Design of RC piled jetty
Week 10	Breakwaters – Introduction, functions, types, failure modes and general design principles
Week 11	Design of breakwaters
Week 12	Final exam / jury
Week 13	-
Week 14	-

Monsoon Semester, 2022, Faculty of Technology, CEPT University

Code	SD4619	Course	Design of Liquid Retaining Structures		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

This course covers the analysis, design and detailing aspects of circular, square and Intze tanks. The behavior of circular and square tanks will be analyzed with respect to its position on ground or underground and also with conditions of empty and full along with the effect of soil pressure and water tables. Serviceability checks will be done based on the elements and the forces acting on them. The complexities will be introduced after that and students will be analyzing those tanks on the basis of the previous lectures of the basic tanks. After the tanks resting on ground and undergrounds, the discussions will be based on elevated tanks. Various types of elevated tanks will be shown and behavior pattern of each will be discussed briefly in class. The problem of Intze tank will be done with that resting on shaft or columns. The relevant staging code and earthquake codes will be discussed. we will also analyze the liquid retaining structures via software to understand the behavior of structures. and the same will be helpful to interpret and compare the software results. The lecture course shall culminate with the analysis, design and detailing of a given problem statement.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Analysis and Design of liquid retaining structures for Water and Sewage treatment plants,

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	History of Liquid retaining structures and understanding the behavior of the same with different parameters and conditions. Understanding the provision given in IS 3370-2021 for different design methodologies i.e. limit state method and working stress method.
Week 2	Analysis and Design of Circular tanks resting on ground and underground. Doing the further analysis by considering seismic parameters. Applying the serviceability checks. Detailing the tanks.
Week 3	Analysis and Design of Circular tanks resting on ground and underground. Doing the further analysis by considering seismic parameters. Applying the serviceability checks. Detailing the tanks.
Week 4	Analysis and Design of Circular tanks resting on ground and underground. Doing the further analysis by considering seismic parameters. Applying the serviceability checks. Detailing the tanks.
Week 5	Analysis and Design of rectangular tanks resting on ground and underground. Doing the further analysis by considering seismic parameters. Applying the serviceability checks. Detailing the tanks.
Week 6	Analysis and Design of rectangular tanks resting on ground and underground. Doing the further analysis by considering seismic parameters. Applying the serviceability checks. Detailing the tanks.

Week 7	INTERNAL EXAM
Week 8	Understanding the behavior of Intze tank resting on shaft and column frames. Analyzing considering earthquake parameters, applying the serviceability checks and detailing the same.
Week 9	Understanding the behavior of Intze tank resting on shaft and column frames. Analyzing considering earthquake parameters, applying the serviceability checks and detailing the same.
Week 10	Understanding the behavior of Intze tank resting on shaft and column frames. Analyzing considering earthquake parameters, applying the serviceability checks and detailing the same.
Week 11	Introduction to Design softwares to analyze liquid retaining structures & result interpretation through softwares
Week 12	Project formulation and discussion
Week 13	Project formulation and discussion
Week 14	FINAL EXAM

Monsoon Semester, 2022, Faculty of Technology, CEPT University

Code	SD4620	Course	Prestressed Concrete		
Typology	Courses	Credits	2	GPA/NonGPA	GPA

Long Description :

Now a days prestressed concrete is a very widely and efficiently used structural forms in large scaled infrastructures projects as well as special building structures. This course is intended to provide the student with the basic tools required to design and build prestressed concrete structures. Emphasis will be placed on the behavior of prestressed concrete under load along with potential failure mechanisms. The important features of the prestressed like, materials, prestressed types, stages of the loadings, various losses of prestressed concrete, Serviceability & Ultimate stage analysis will be covered during the course. The effect of the cable profile & eccentric prestressed force will be discussed and critical thinking on the suitable adoption of the cable profile will be explored. Each topic will be elaborated properly with well formatted PowerPoint presentations, supplementary resource material and assignments to ensure proper conceptualization of topics. A full scale actual field problem of prestressed concrete design will be solved with step by step calculations. Design criteria of Serviceability Limit State and Ultimate Limit State both will be covered.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Conceptual understanding of prestressed concrete
- Behavior, analysis, design & detailing of prestressed concrete member
- Exposure to actual industry practice to design & detail prestressed concrete structures
- Able to resolve any filed query or design/ construction related problem arises during construction

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Introduction: historical development, fundamentals, types of prestress, materials & hardware, difference between RCC & prestress, advantages & dis-advantages of Prestress
Week 2	Various materials and hardware required, methods of executing prestress concrete
Week 3	Losses of Prestress : Short term & Long term Losses. Basic understanding of the prestress losses (why & How), various factors affecting the losses
Week 4	Losses of Prestress : Short term & Long term Losses. Basic understanding of the prestress losses (why & How), various factors affecting the losses
Week 5	Losses of Prestress : calculations of losses
Week 6	Step by step procedure of Prestressed beam design, Preliminary Prestress, to fix up numbers of cables/strand required, to fix up cable profile

Week 7	SLS Stress check of the prestressed members at various stages of the loadings, permissible stresses in concrete during transfer and service stage
Week 8	Ultimate flexural design of prestressed member
Week 9	ULS shear design, effect of the prestress in shear design
Week 10	ULS shear design, effect of the prestress in shear design
Week 11	Cable elongations, Design of End Block
Week 12	Detailing of a prestressed beam
Week 13	nil
Week 14	nil

Winter Semester, 2022, Faculty of Design, CEPT University

Code	W22FD020	Course	LEARN THE ART AND CRAFT OF MOBILE FILM-MAKING		
Typology	-	Credits	3	GPA/NonGPA	GPA

Long Description :

The course will introduce students to mobile film-making basics. They will learn how to shoot the film with mobile phone and how to use gears to enhance film outputs. The three major building blocks of films: pre-production, production and post-production will be taught. The participants will be learning how to visualize ideas, conceptualise and represent through the medium of films. It will teach them to write conceptual and executable scripts and screenplays. The logic and techniques to write a gripping and engaging short film will be taught. It will also be in introduction to film-production, shoot scheduling, shot division, lighting the shots, film sound recording and color correction of films. The basics of mobile handling, introduction to sound capturing techniques and sound sync will be demonstrated. The set design and art direction basics will be taught. Basic film editing will be taught. The entire course work will be an hands-on experience and short films are expected to be made as an output by the participants and guided in the end to end process with each passing sessions.

Learning Outcomes :

After completing the Courses, the student will be able to :

- The participants will learn how to write for films. The participants will make a short film at the end of the course.

Daily Plan & Assessment :

Days	Exercise Description
Day 1	Introduction to mobile film making
Day 2	Learning to write story and script
Day 3	Learning the write screenplay
Day 4	Learning shooting techniques
Day 5	Shooting the film
Day 6	Shooting the film
Day 7	Shooting the film with review
Day 8	Learning to edit the film

Spring Semester, 2023, Faculty of Technology, CEPT University

Code	SD4005	Course	Design of Bridges		
Typology	Studio	Credits	14	GPA/NonGPA	GPA

Long Description :

This studio is proposed on the main objective to prepare structural engineers having strong fundamentals, which enables the candidates to have in-depth understanding & knowledge of theory and applications of various engineering aspects. This studio helps to attain desired level of proficiency required for design of bridges. The studio will commence with learning various types of bridges across world with respect to materials and structural system. Factors affecting the selection of bridge site, material, span and system will be discussed elaborately so the students can adapt a suitable configuration for the problem they are assigned with. The shapes and functions of various bridge components such as longitudinal girder, deck slab, diaphragms, crash barriers, parapet, footpath, bearings, pier cap, pier & foundations shall be discussed. The industrial methodology of preparing general arrangement drawing and consideration of loads, impact factor, braking force, wind, seismic shall be discussed. Subsequent exercises will be given to the students to make them reach the final output of the studio. Apart from general discussions of the main components of the bridges, especially for super structure and sub structure, students will be enlightened with the specific Indian standards such as IRC 5, IRC 6 and IRC 112 which are related to necessary clauses for structural design. The grillage analysis that is done in practice will be verified with manual calculations that are required for the quick checking of the structural design, will be dealt with in this studio. The final output of the studio will be in terms of the design conceptualization, optimization, manual calculations, software model and neat structural drawings.

Learning Outcomes :

After completing the Courses, the student will be able to :

- Prepare the General arrangement of bridge as per industrial requirement - Apply the criterion for the suitability/feasibility of the various types of the superstructure. Analysis the various types of bridge superstructure & sub structures - Apply the codal regulations for bridge loadings as per IRC-06, design methodology as per IRC 112 - Report formulation

Weekly Plan & Assessment :

Weeks	Exercise Description
Week 1	Basics of the Bridge Engineering: History & Overview Bridge Definition, Types of Bridge, Bridge Components & its Function, Types of Superstructure & its Suitability, Formulation of Carriageway, Vertical Clearances, Substructure, Foundation, Force Flow
Week 2	Basics of the Bridge Engineering: History & Overview Bridge Definition, Types of Bridge, Bridge Components & its Function, Types of Superstructure & its Suitability, Formulation of Carriageway, Vertical Clearances, Substructure, Foundation, Force Flow
Week 3	General Arrangement of Bridge & Load definitions as per IRC -6
Week 4	Analysis of Bridge Superstructure Preparation Grillage analysis with STAAD software, perform analysis & abstract of results required for design of superstructure

Week 5	Analysis of Bridge Superstructure Preparation Grillage analysis with STAAD software, perform analysis & abstract of results required for design of superstructure
Week 6	Analysis of Bridge Superstructure Preparation Grillage analysis with STAAD software, perform analysis & abstract of results required for design of superstructure
Week 7	Mid semester exam/jury
Week 8	Design of Superstructure Design of Longitudinal girder/webs/slabs, deck slab, diaphragms. checking adequacy of the various dimensions, computation of flexural reinforcement/ Prestressing steel some basic short design calculations to be carry out with hand calculations and hand sketches
Week 9	Design of Superstructure Design of Longitudinal girder/webs/slabs, deck slab, diaphragms. checking adequacy of the various dimensions, computation of flexural reinforcement/ Prestressing steel some basic short design calculations to be carry out with hand calculations and hand sketches
Week 10	Design of Superstructure Design of Longitudinal girder/webs/slabs, deck slab, diaphragms. checking adequacy of the various dimensions, computation of flexural reinforcement/ Prestressing steel some basic short design calculations to be carry out with hand calculations and hand sketches
Week 11	Analysis & design of Substructures
Week 12	Analysis & design of Substructures
Week 13	Detailing of Super Structure & Substructures
Week 14	Detailing of Super Structure & Substructures
Week 15	Final exam/jury/Portfolio
Week 16	Final exam/jury/Portfolio